

Analysis of the CoAP data within a DDOS attack

Project Presentation

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2026-07-15

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Intro

- starting at the 25th of August 2025 there was a DDOS attack on mobi3
- part of that multi-protocol DDOS attack was CoAP traffic
- CoAP is also the focus of my work at RIOT and in other modules

- based upon the IRTF draft: *Amplification Attacks Using the Constrained Application Protocol (CoAP)* [1]
- are a type of reflection-based volumetric DDoS/DoS attacks

- Simple Amplification Attacks
- Amplification Attacks using the Observe Option
- Amplification Attacks using Group Requests
- Amplification Attacks using Group Requests and the Observe Option

Are the 4 types of amplification attacks recognizable in the DDOS attack on mobi3?

- 779 GiB of data: *net/archive/ddos_data*
- only a subset of data was used (next slide)
- parsed with *tshark*
- *general_data_set* and *coap_data_set*


```
All packets      : 696,340,178
All bytes        : 801,727,874,184 (801727.9 MB)
Timepoints       : 2025-08-13 15:51:49.573544 → 2025-08-13 17:46:30.464776
Duration         : 6880.9s / 114.7min / 1.9hours
Ø Packetrate     : 101,199.1 pkt/s
Ø Bitrate        : 932.1 Mbit/s
Average packet size : 1151.3 bytes
Maximum packet size : 18890 bytes
```

Figure 1: `_general_data_set_`

```
CoAP distribution      : 0.07%
All packets           : 696,340,178
CoAP packets          : 51,418,885
All bytes              : 801,727,874,184 (801727.9 MB)
CoAP bytes            : 68,581,946,168 (68581.9 MB)
Ø CoAP Packetrate     : 101,199.1 pkt/s
Ø CoAP Bitrate        : 932.1 Mbit/s
Average packet size   : 1333.8 bytes
Maximum packet size   : 1514 bytes
```

Figure 2: _coap_data_set_

Plots

```
All packets      : 696,340,178
All bytes        : 801,727,874,184 (801727.9 MB)
Timepoints       : 2025-08-13 15:51:49.573544 → 2025-08-13 17:46:30.464776
Duration         : 6880.9s / 114.7min / 1.9hours
Ø Packetrate     : 101,199.1 pkt/s
Ø Bitrate        : 932.1 Mbit/s
Average packet size : 1151.3 bytes
Maximum packet size : 18890 bytes
```

Figure 3: Overview of analyzed data

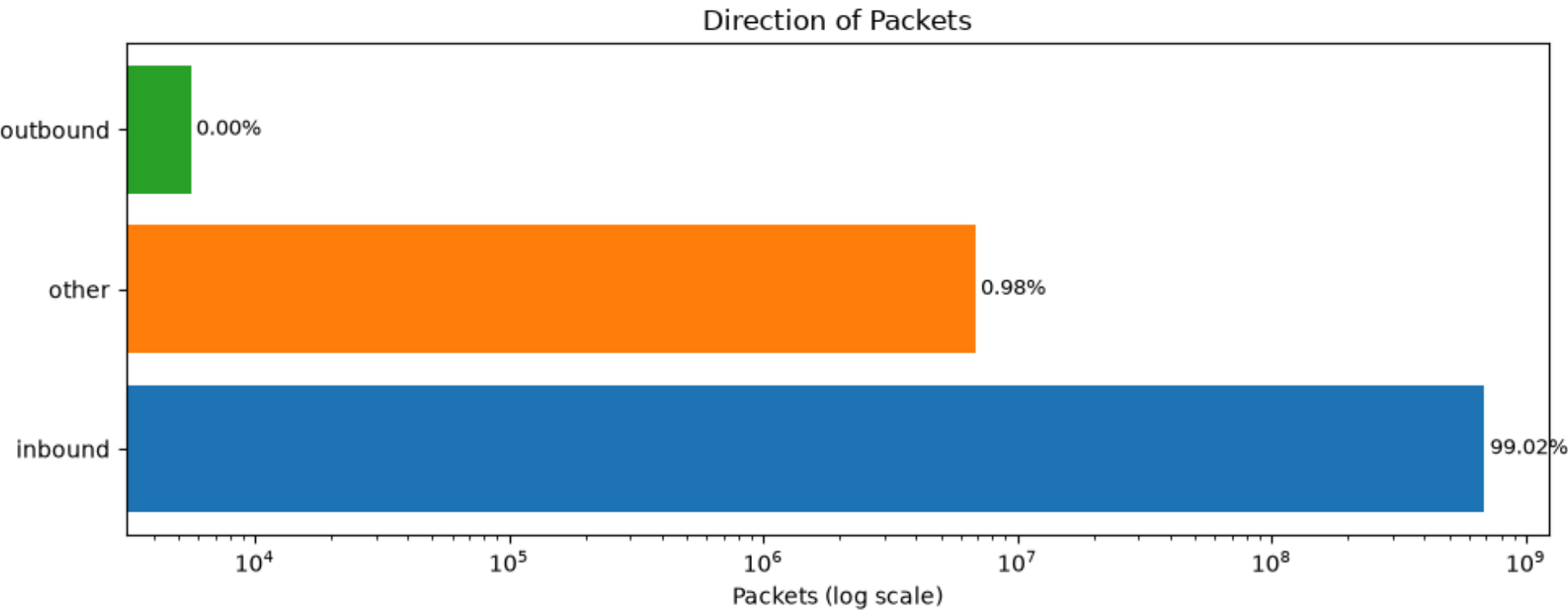


Figure 4: Share of Packet Directions

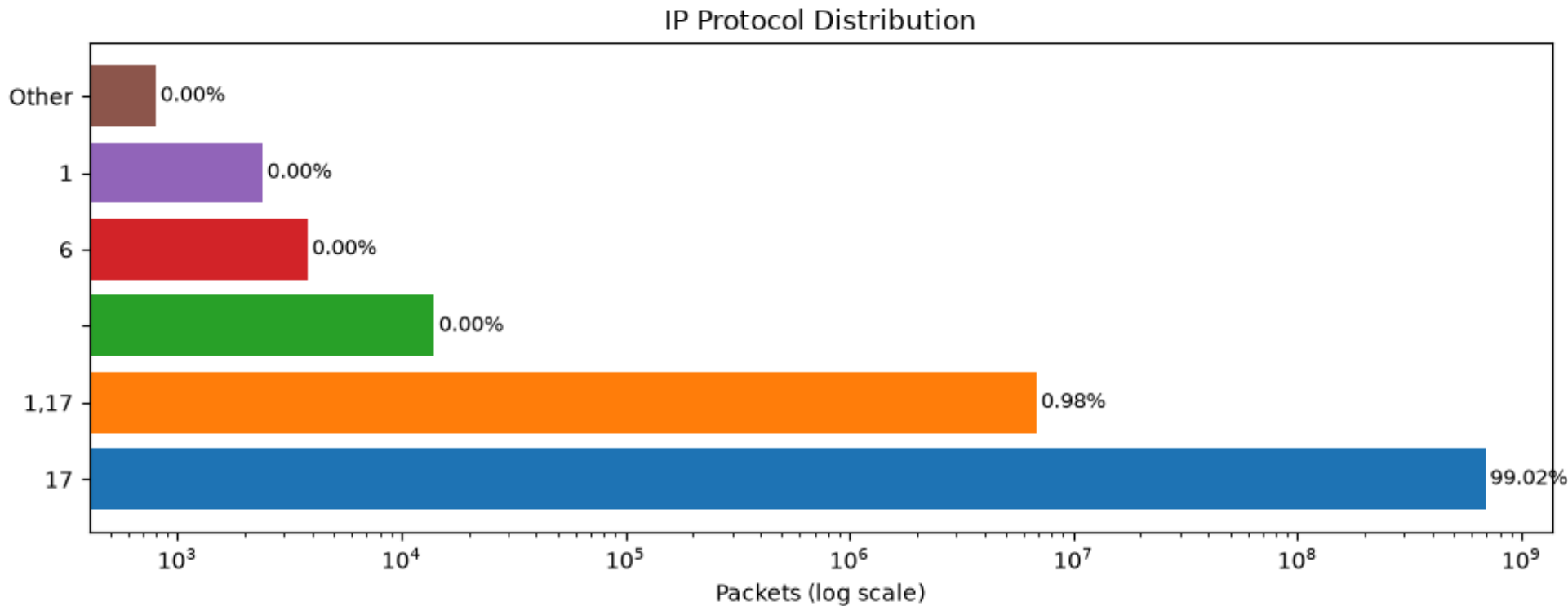


Figure 5: Share of IP Protocols

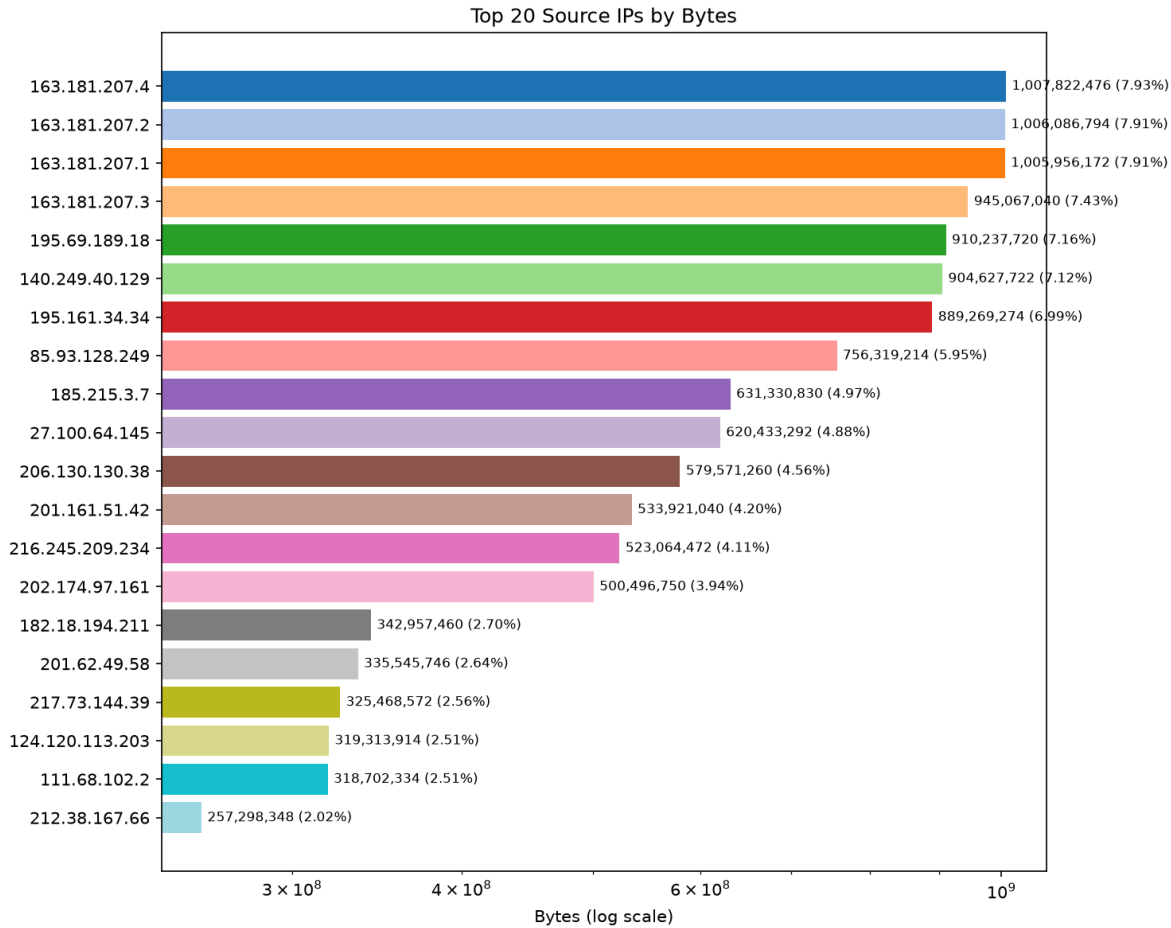


Figure 6: Top Talkers

```
CoAP distribution      : 0.07%
All packets           : 696,340,178
CoAP packets           : 51,418,885
All bytes              : 801,727,874,184 (801727.9 MB)
CoAP bytes             : 68,581,946,168 (68581.9 MB)
Ø CoAP Packetrate     : 101,199.1 pkt/s
Ø CoAP Bitrate        : 932.1 Mbit/s
Average packet size   : 1333.8 bytes
Maximum packet size   : 1514 bytes
```

Figure 7: Overview of CoAP data

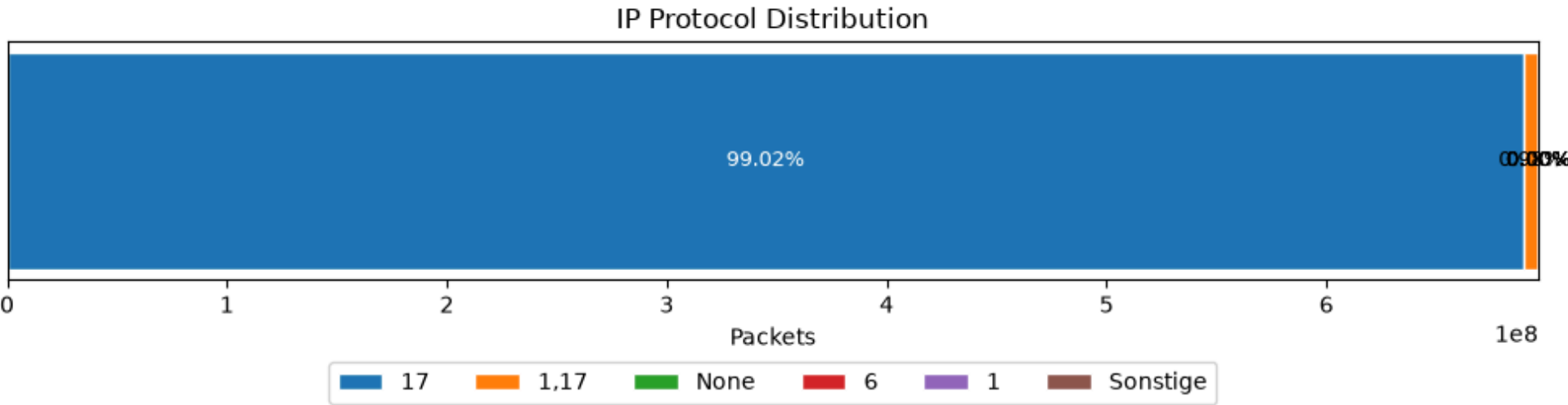


Figure 8: Share of IP Protocols of CoAP packets

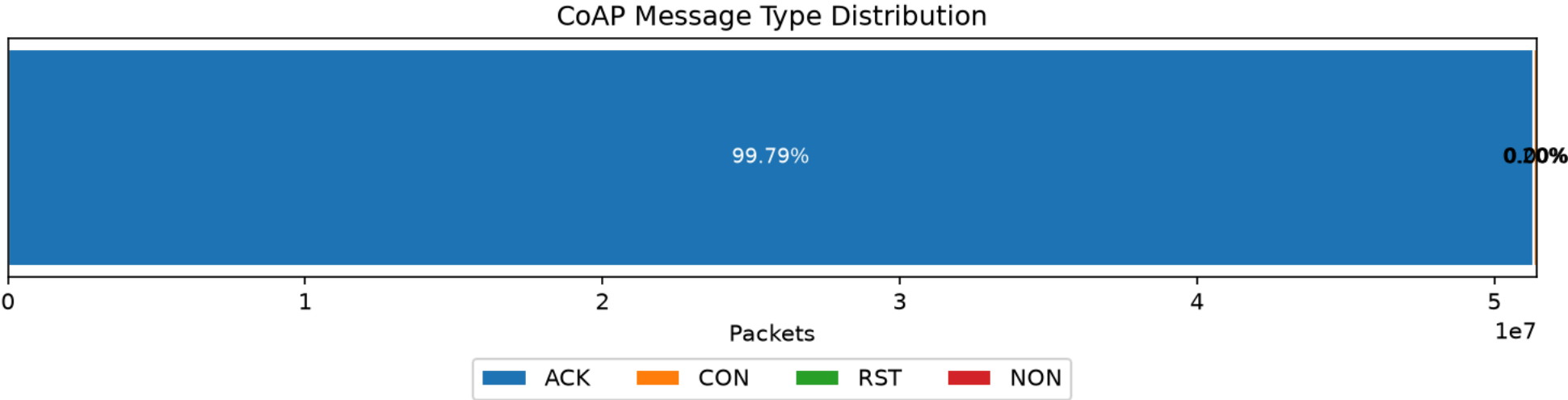


Figure 9: Share of CoAP Message Types

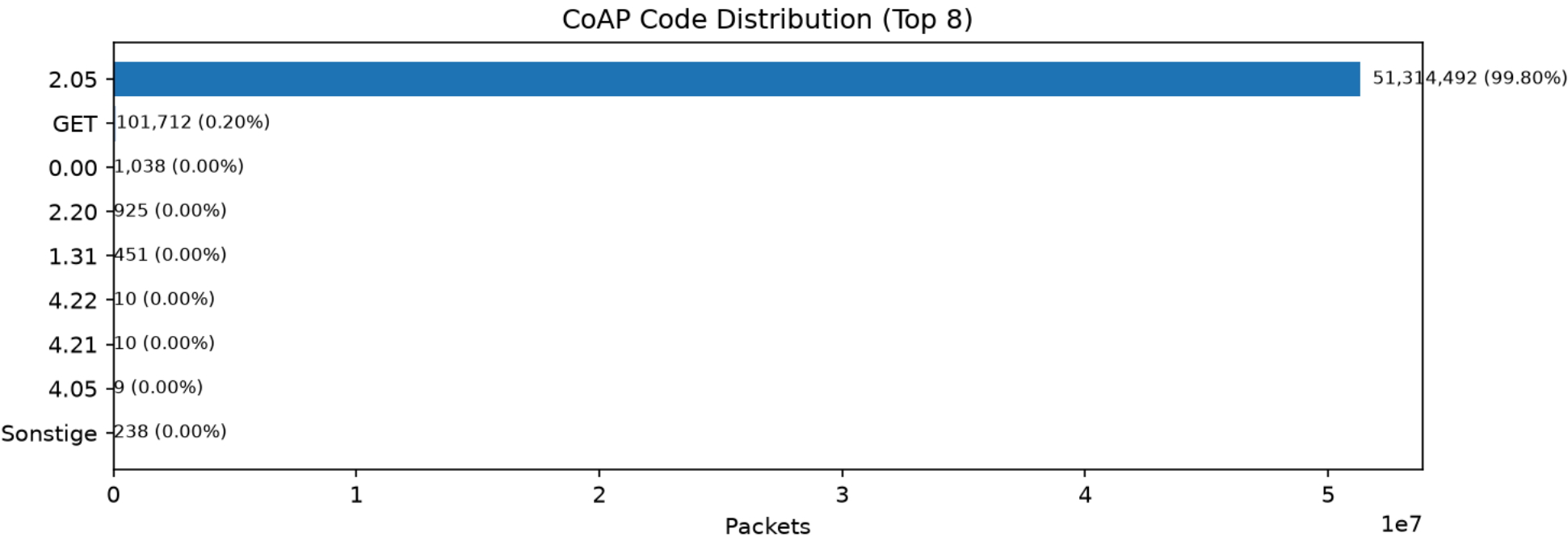


Figure 10: Share of CoAP Codes

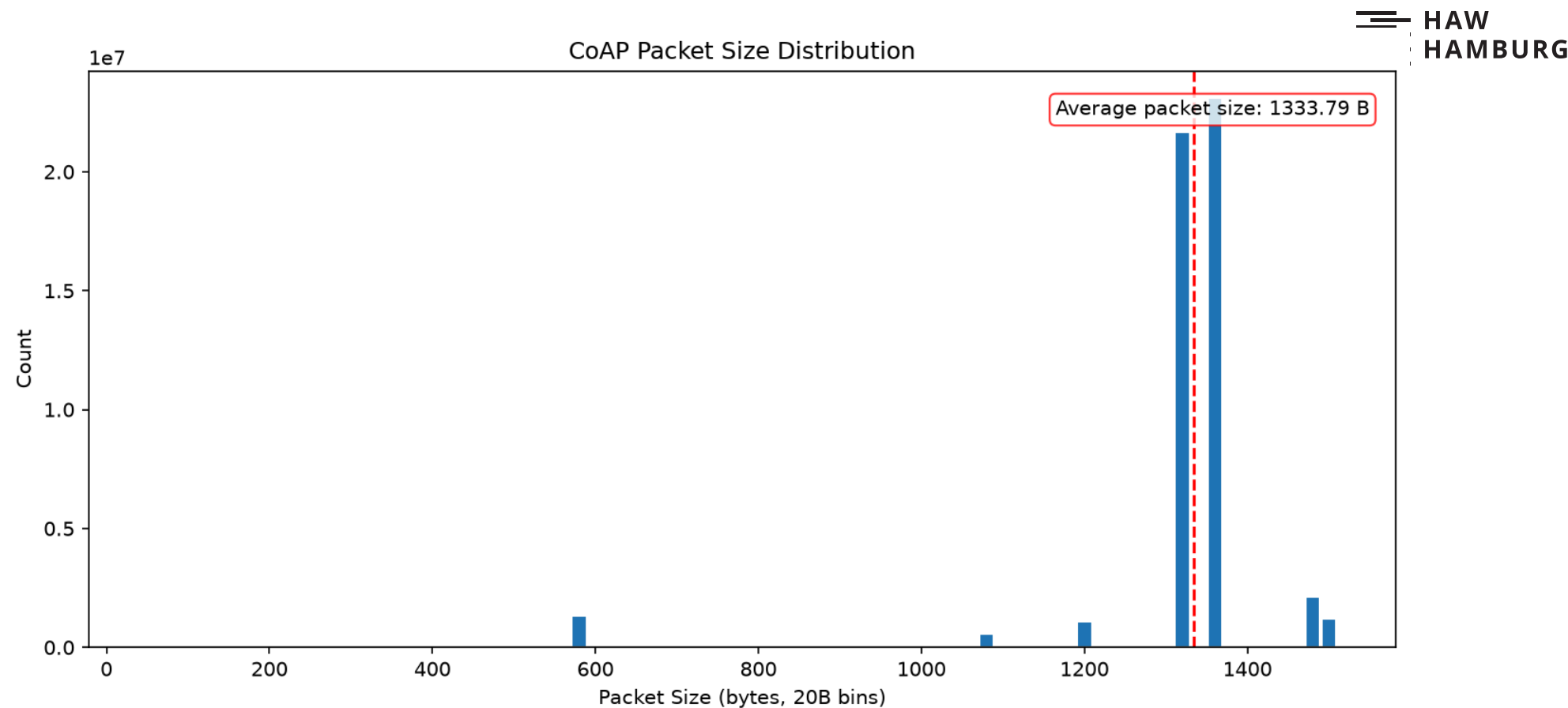


Figure 11: Packet Sizes within CoAP

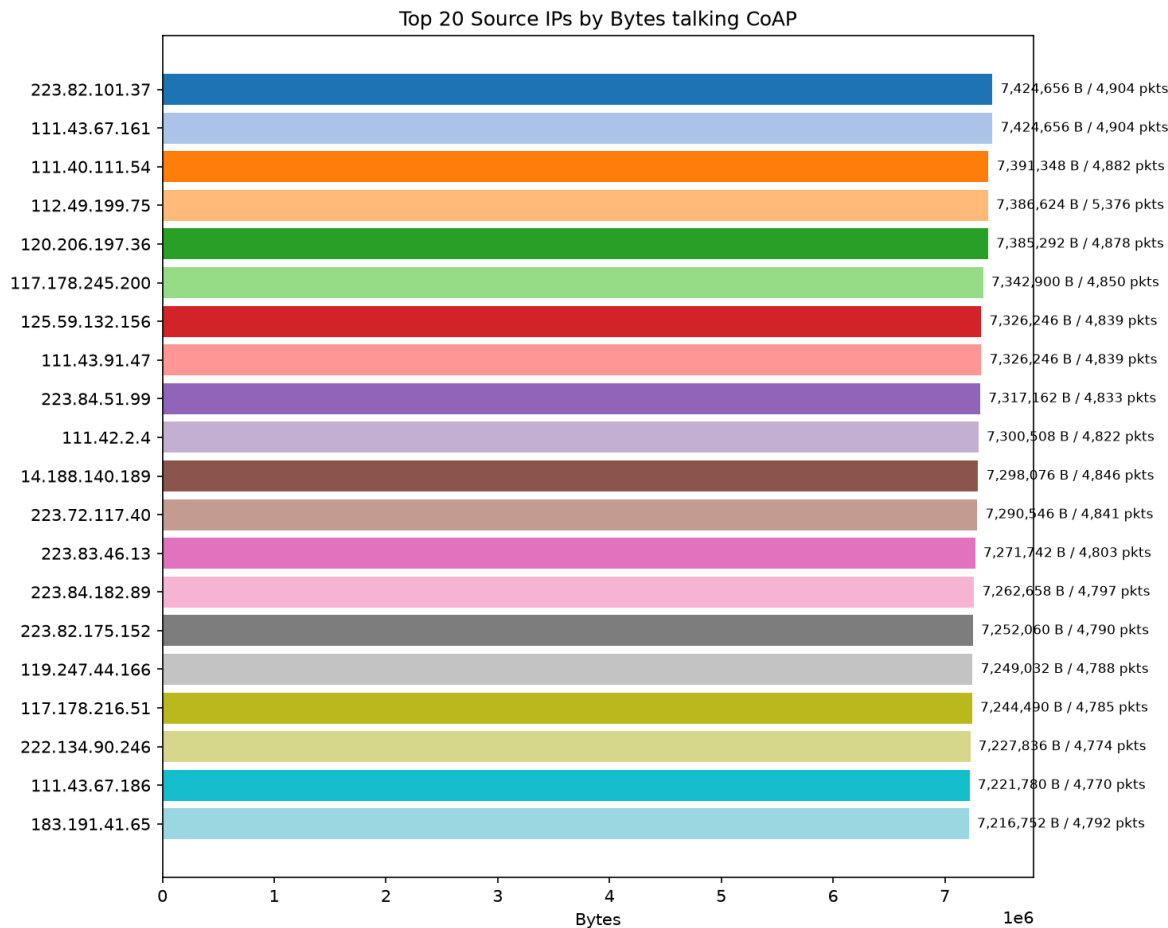


Figure 12: Top Talkers of CoAP

Outro

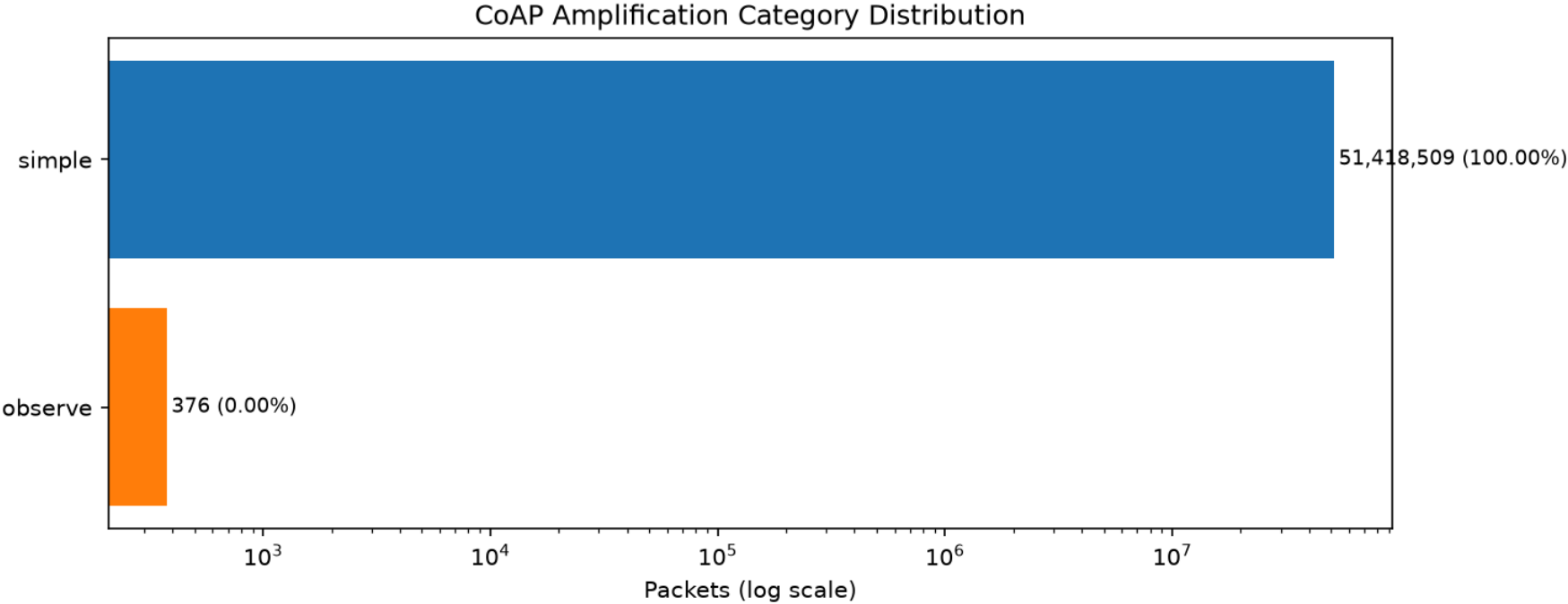


Figure 13: Amplification Attacks Categories

Yes, but in the attack only Simple
and Observe were used!

- only around 0.7% of the DDOS attack packets consists of CoAP packets
- as expected is most of the data transported by UDP
- as expected are most of CoAP packets of method GET in a message of type ACK
- no DTLS was found

- [1] J. P. Mattsson, G. Selander, and C. Amsüss, “Amplification Attacks Using the Constrained Application Protocol (CoAP),” Internet Engineering Task Force, Internet-Draft draft-irtf-t2trg-amplification-attacks-05, Jun. 2025. [Online]. Available: <https://datatracker.ietf.org/doc/draft-irtf-t2trg-amplification-attacks/05/>